(3) $f(\theta) = -2\sin(\theta) - \theta^{10} - 12, \quad \theta_0 = \pi/4$

$f(\theta_0) = f(\pi/4) = -2\sin(\pi/4) - (\pi/4)^{10} - 12$

$= -1.414 - 0.0893 - 12$

$= -13.5$

$f'(\theta) = -2\cos(\theta) - 10\theta^9$

$f'(\pi/4) = -1.414 - 1.137 = -2.55$

$L(\theta) = f(\theta_0) + f'(\theta_0)(\theta - \theta_0)$

$= -13.5 + -3.14(\theta - \pi/4)$

$L(\theta) = -3.14\theta + 2.55$

$= -13.5 + -2.55(\theta - \pi/4)$

$\boxed{L(\theta) = -2.55\theta - 11.5}$